

Response to reviewers

R1

General comments

General comment 1

We agree that the comparison between the two systems is not perfect. The two other differences besides sea-ice initial conditions are the ocean initial conditions and the ensemble generation.

Ideally we would compare the ocean initial conditions – Southern Ocean SSTs in particular and recreate a time-lagged ensemble with ACCESS-S1. Unfortunately, the raw ACCESS-S1 hindcast is no longer available, so we cannot do either.

However, on their initial assessment of ACCESS-S2, Wedd et al. (2022) suggests that initial conditions in the Southern Ocean are similar between models, so we think that this should be a minor difference. The ensemble generation would mainly affect ensemble spread, which is comparable between systems after just a few days.

We added a section pointing out these limitations.

General comment 2

In the revised manuscript we have revised this section to improve clarity (lines XX). We have also added a figure showing the correlation between sea-ice thickness and sea-ice concentration amplitude (Fig. XX), which supports our assertion that thinner sea ice is more sensitive than thicker sea ice.

General comment 3

Since we compute anomalies relative to each model's own lead-time-dependent climatology, this applies a first-order bias correction.

Specific comments

Specific comment 1

The general definition of skill score in section 4.3 of Murphy and Daan (1985) states that it can be derived from any error measure. Section 5.3.2 in gives examples of skill scores in terms of Mean Absolute Deviation and Mean Squared Error, but Root Mean Square Error is also a valid measurement from which to derive skill scores.

Specific comment 2

We have corrected the error.

Specific comment 3

That is true. However, in those months RMSE increases with lead time. We added Line XX: “This effect is seen in all months except for July to September.” and a supplementary figure showing RMSE as a function of lead time for each month.

Specific comment 4

During data assimilation, if sea-ice concentration innovations are positive, these are added to the first ice category with a fixed thickness of 50 cm. If they are negative, they are removed from the thinnest category first and from thicker categories if needed. Therefore, although this system only assimilates sea-ice concentration data, sea-ice thickness is noneless modified.

We have added this clarification to the revised manuscript.

Specific comment 5

See response to General comment 1.

Specific comment 6

We have corrected this.

Specific comment 7

We have corrected this.

Specific comment 8

Yes. This is now corrected. We have modified our discussion about Libera et al. (2022). See response to R3.

Specific comment 9

We are not aware of any objective criterion to derive propagation of error. We follow Holland et al. (2013), who based their conclusions on visual inspection of a similar Hovmoller plot.

Specific comment 10

We have corrected this.

Figures

Figure 1 The triangles show the mean forecasted extent for the first of the month using the maximum lead time possible. That lead time is always about 7 months, but the exact number of days depends on the month due to the different number of days in each month. We have now clarified this in the caption.

Figure 5 The numbers in brackets represent the 95% confidence interval. We have now clarified this in the caption. All correlations are statistically different from zero, however that comparison implies a constant forecast as the “null hypothesis”, which is not very informative. A better comparison would be a naive forecast of persistence, and we perform that calculation in the next section.

Figure 6 We assume a normal distribution for sea-ice extent, so the standard deviation has a chi-squared distribution with $N - 1$ degrees of freedom (where N is the number of samples). We then divide all the values by the observation standard deviation. We have modified the caption to make this clearer.

Figure 9: With our data, it is not possible to say to what extent the oceanic and atmospheric data assimilation improve sea-ice forecasting. It might very well be the case that if it were not for data assimilation, the forecasts would have been even worse as it is evident that some information is being transferred from the atmosphere or ocean to the ice in some months (Figure 1).

Figure 10-11 We have added stippling showing statistical significance.

Additional references We thank you for the recommendations. Xiu et al. (2025) was particularly useful since it uses a similar methodology.

R2

General comments

Improper methodology for addressing the stated objective

You are correct in pointing out that there are other differences between the two forecasting systems. The two other differences besides sea-ice initial conditions are the ocean initial conditions and the ensemble generation.

Ideally we would compare the ocean initial conditions – Southern Ocean SSTs in particular and recreate a time-lagged ensemble with ACCESS-S1. Unfortunately, the raw ACCESS-S1 hindcast is no longer available, so we cannot do either.

However, on their initial assessment of ACCESS-S2, Wedd et al. (2022) suggests that initial conditions in the Southern Ocean are similar between models, so we think that this should be a minor difference. The ensemble generation would mainly affect ensemble spread, which is comparable between systems after just a few days.

We added a section pointing out these limitations.

Quality of figures

We improved the figures based on reviewer feedback.

Specific comments

L64, Section 2.1 We modified the section title and made it clear that ocean and sea-ice initial conditions are the only differences between ACCESS-S1 and ACCESS-S2.

L125, Section 2.3 We added bias and correlation to the Methods section. We added confidence intervals when possible (Figures 5, 6, 8, 9, 14) and significant p-values to Figures 11-13.

L126–127 The hindcast ensemble mean sea-ice extent is defined as the mean sea-ice extent of individual ensemble members. We added this clarification in the Methods section.

Figures 3 and 4 We changed the colour scale and added lat-lon lines.

L194 We clarified that we are talking about RMSE.

L214–215 There was an error in the labeling of the plots where observations were labeled as ACCESS-S2. This is now fixed.

L245 Fixed

Figures 9 and 10 We have moved Figure 10 to the supplementary material, and replaced it with a figure highlighting the lead time at which ACCESS-S1 and ACCESS-S2 forecasts are statistically indistinguishable.

Figures 11–13 We altered the colour scale, added a vertical grid, and smoothed values to increase readability.

Caption of Figure 14 We have added “days” in the figure to make the unit clearer.

L277–278 Yes, the spread is slightly higher in S1 between Feb and April. Lines XXX now point that out.

L279–280 and Figure 14 We have reworked this figure and have removed the decomposition of variance information based on reviewer comments.

L307–308 Lines XX now read: The way sea-ice thickness is handled is that sea-ice concentration innovations are either added to the first ice category with a fixed thickness of 50 cm or removed from the thinnest category first and from thicker categories if needed.

R3

Major comments

Major comment 1

We have made it clear that all computations were done using only the overlapping period between hindcasts.

Major comment 2

We appreciate the reviewers concern about monthly subplots. However, to show the evolution of results over the year and for consistency across all figures we have decided to keep monthly subplots.

Major comment 3

We have reworked this part of the manuscript to improve clarity. We have also added a Supplementary Figure showing the correlation between sea-ice thickness and sea-ice concentration amplitude (Fig. XXX), which supports the assertion that thinner sea ice is more sensitive than thicker sea ice.

Major comment 4

We thank the reviewer for this comment. We have thought about the discrepancy further and come to the conclusion that there are many differences between our work and Libera et al. (2022) that actually prevent a direct comparison.

First, by analysing sea-ice area, Libera et al. (2022)’s results are insensitive to the distribution of sea ice within the Weddell Sea region, while our use of RMSE of sea ice concentration anomalies takes this spatial distribution into account. Figure 1 illustrates how the choice of method can lead to different results.

Second, Libera et al. (2022) used lagged autocorrelation as the definition of predictability – essentially measuring persistence – while we are analysing predictions from a fully coupled model.

We have decided to change our wording to make the distinction clearer.

Major comment 5

You are absolutely right. We have added a figure and a paragraph highlighting the key result.

Major comment 6

Figures 11–13 are information-dense but low in informativeness, with cumbersome correspondence between text and subplots. Alternative visualization methods (e.g., spatial aggregation, key region highlighting, or summary statistics) are recommended to convey core messages (e.g., regional skill differences) more directly.

Minor comments

1. We clarified that that result is relevant for the Arctic.
2. We changed the section titles.
3. It is not uncommon to use the cold colour to denote higher sea ice concentrations (which generally imply colder temperature) and the warm colour to denote lower sea ice concentrations (associated with higher temperatures) (e.g. Espinosa, Blanchard-Wrigglesworth, and Bitz 2024; Massonnet et al. 2023; Morioka et al. 2022). We prefer to use this convention.
4. We have changed the column order.
5. We have added units to that and other figures.

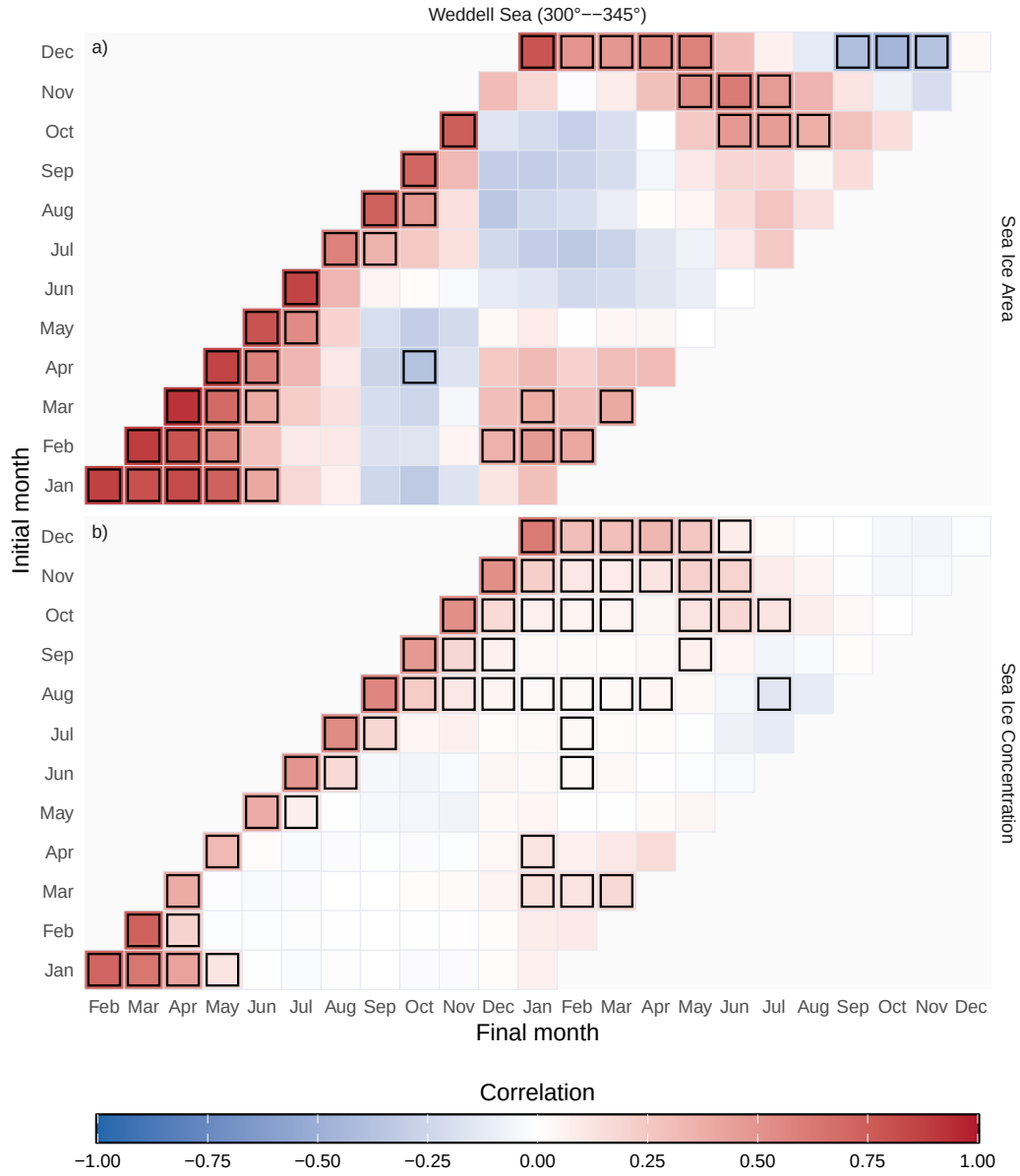


Figure 1: Autocorrelation of monthly sea ice area at different lags (panel a) and mean lagged sea ice concentration anomaly pattern correlation at different lags (panel b) in the Weddell Sea. Values statistically different from zero at the 0.05 level are highlighted with black edges.

6. As with warm/cold colours used to represent sea-ice anomalies, different publications adopt different conventions of how to spell “sea ice”. We chose to follow [Merriam-Webster’s guidelines](#) and use “sea ice” as an open compound noun when used on its own (“Accurately modelling Antarctic sea ice is essential...”) and “sea-ice” as a hyphenated adjective (“...progress in Antarctic sea-ice forecasting system has lagged behind...”). Both terms can be found in the literature (eg. Raphael 2003; Morioka et al. 2022) and are in common usage (Figure 2).

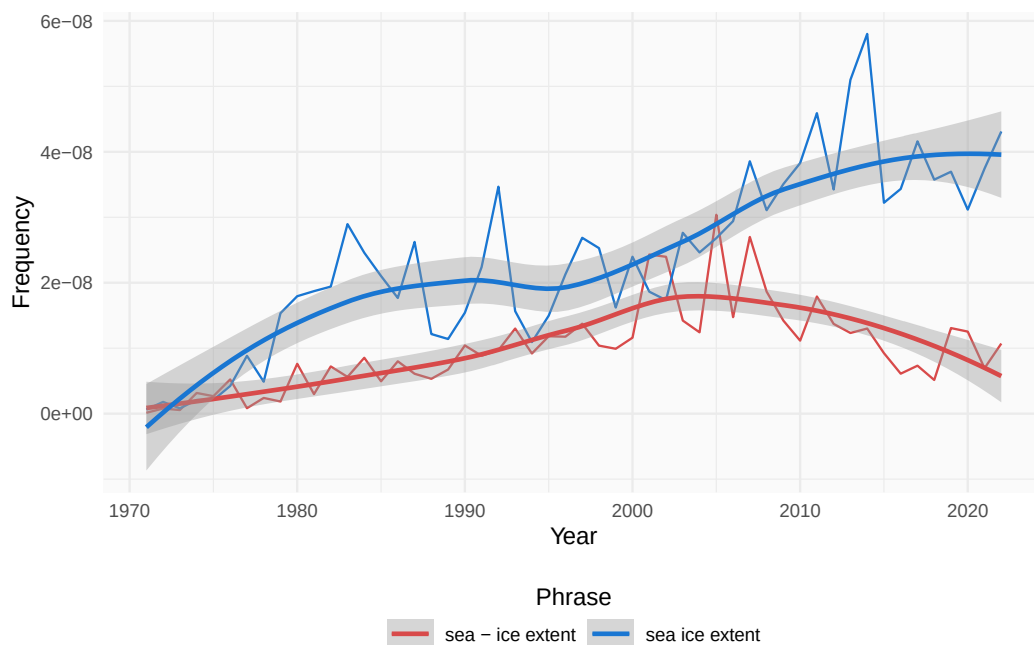


Figure 2: Frequency of the phrases “sea-ice extent” and “sea ice extent” from 1970 according to Google’s Ngram viewer.

7. We have corrected the caption
8. We have removed this panel in the revised version.
9. We have moved the conclusions to its own section instead of a subsection.

References

- Espinosa, Zachary I., Edward Blanchard-Wrigglesworth, and Cecilia M. Bitz. 2024. “Understanding the Drivers and Predictability of Record Low Antarctic Sea Ice in Austral Winter 2023.” *Communications Earth & Environment* 5 (1): 1–9. <https://doi.org/10.1038/s43247-024-01772-2>.

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- Libera, Stephy, Will Hobbs, Andreas Klocker, Amelie Meyer, and Richard Matear. 2022. “Ocean-Sea Ice Processes and Their Role in Multi-Month Predictability of Antarctic Sea Ice.” *Geophysical Research Letters* 49 (8): e2021GL097047. <https://doi.org/10.1029/2021GL097047>.
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- Raphael, M. N. 2003. “Impact of Observed Sea-Ice Concentration on the Southern Hemisphere Extratropical Atmospheric Circulation in Summer.” *Journal of Geophysical Research: Atmospheres* 108 (D22). <https://doi.org/10.1029/2002JD003308>.
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- Xiu, Yongwu, Yiguo Wang, Hao Luo, Lilian Garcia-Oliva, and Qinghua Yang. 2025. “Impact of Ocean, Sea Ice or Atmosphere Initialization on Seasonal Prediction of Regional Antarctic Sea Ice.” *Journal of Advances in Modeling Earth Systems* 17 (2): e2024MS004382. <https://doi.org/10.1029/2024MS004382>.